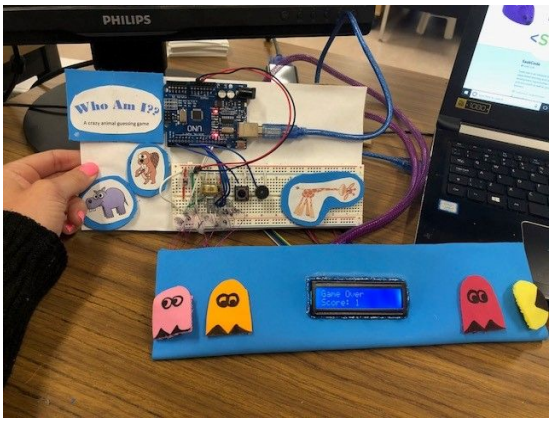
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Robot Name: Arduino

Project Four: Display

Activity 4D: "Who Am I?" Game

Grade Level	7-8	**Difficulty**	Beginner
Subject	Practical and Applied Arts Science		
Description	"Who Am I?" is based on the popular Hedbanz game or HeadsUp! app. It's a fun party game in which a player holds an LCD screen to his/her forehead, and other players give hints to help the player with the LCD guess the word on the screen. 		
Outcome(s)	PAA - Electronics Modules ELEC02A, ELEC04A - Robotics TBA Science - CE9.1, CE9.2		
Required Prior Learning	Activity 1 Getting Started with Arduino Activities 4A, 4B, 4C		
Time Frame	Two hours		
Materials	LCD screen, push button, piezo buzzer, 20 jumper wires, potentiometer, arduino, breadboard, USB cable Worksheet Activity 4D: Who Am I? Game		
Additional Resources	Video of the game- https://www.youtube.com/watch?v=UpBm3x8iI0E Guide To Installing Arduino Software, Easy Intro to BlocklyDuino & C		
Assessment	Review lesson ending questions with students for understanding and future application.		

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Activity 4D: “Who Am I?” Game

Procedure

Review the organization of the kit and how groups need to be careful with all the components, how they are used, and how they need to be kept organized and put back into their place when done.

Divide your class into groups of two (Paired Programming). Paired programming is highly recommended when working with coding and robotics so that students can piggyback off each others ideas and strengths.

Students are going to build onto the circuit from Activity 4A: Using an LCD screen. Students will add on a buzzer and a button. Hand out the student handout- Activity 4C- “Who Am I?”, and guide students through the instructions. Once students have wired their Arduino, students will then copy and paste the provided code into the Arduino Editor and download it onto the Arduino Board. Once downloaded, the students will be able to play the game and later alter the game.

Student Handouts

Activity 4C - “Who Am I?” Game

Assessment Guide

Circulate the room observing and assisting with students so that they can complete the project. Correct the assignment sheet as a class ensuring that students share their errors and successes while reinforcing the importance of persistence and patience in programming.

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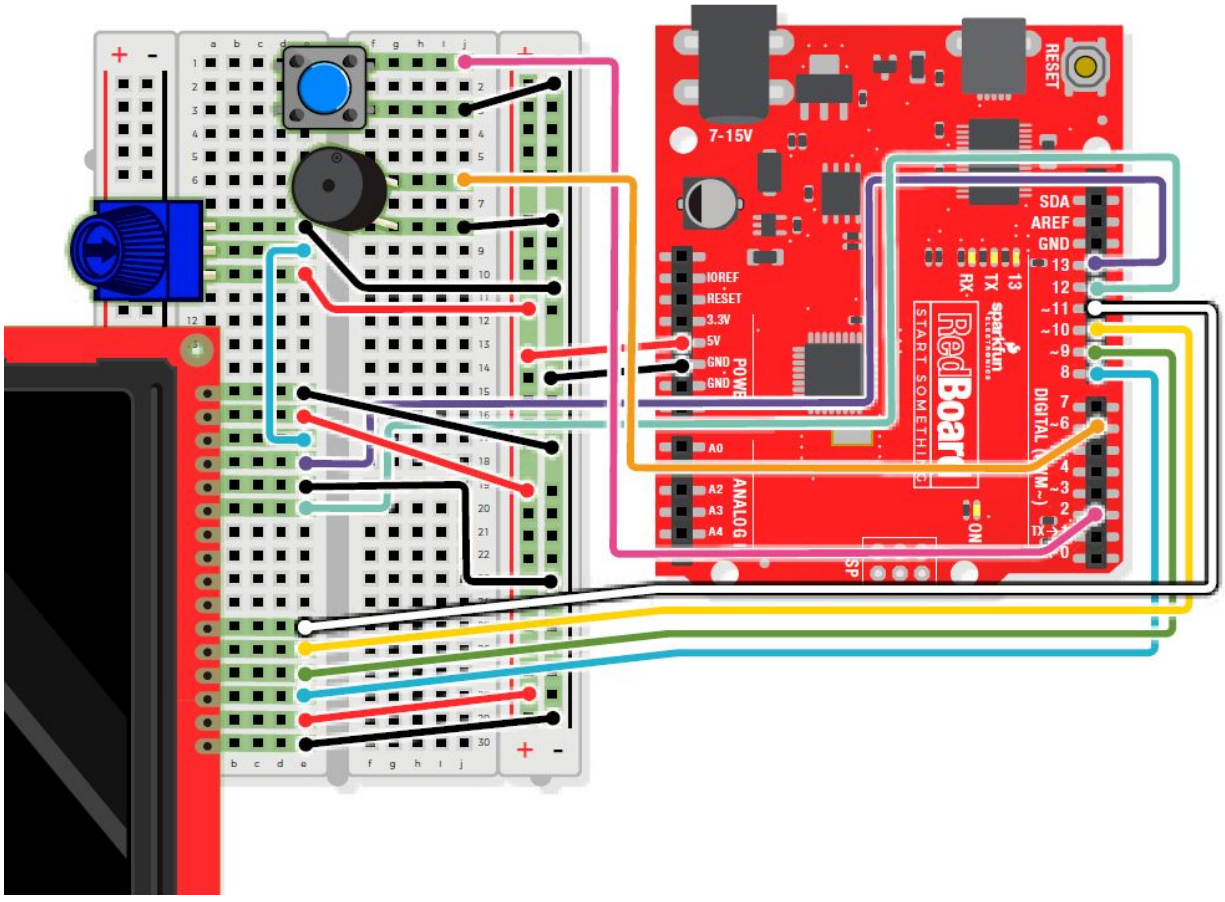
Activity 4D: "Who Am I?" Game

Purpose: To learn how to wire a circuit that includes a potentiometer, buzzer, a button and an LCD display screen.

Materials: LCD screen, push button, piezo buzzer, 20 jumper wires, potentiometer, Arduino Uno board, breadboard, USB cable

Procedure:

1. Wire the project using the following schematic.



2. Log into your arduino account so that you can save your sketch.

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3. You can create your code in the BlocklyDuino Editor or at the BlocklyDuino Website, then paste it to the Arduino IDE. Simply “drag” code blocks to create your code. Now select the “Arduino” tab you will see the code on the right.

4. Copy the Arduino Code from BlocklyDuino into the Arduino Editor  . To paste the code into your Arduino Editor be sure to use ctrl “v.” Hook up your arduino to your computer using the blue USB cable.

OR

Copy and paste the following code into the Arduino Editor:

```
#include <LiquidCrystal_I2C.h>           //the liquid crystal library contains commands for printing to the display
LiquidCrystal_I2C lcd (0x27, 16,2);     // tell the RedBoard what pins are connected to the display

int buttonPin = 2;                  //pin that the button is connected to
int buzzerPin = 6;                  //pin for driving the buzzer
int buttonPressTime = 0;            //variable to show how much time the player has left to guess the word (and press the button)

long timeLimit = 15000;             //time limit for the playe to guess each word
long startTime = 0;                 //used to measure time that has passed for each word
int roundNumber = 0;                    //keeps track of the roundNumber so that it can be displayed at the end of the game
const int arraySize = 25;

const char* words[arraySize] = {"moose", "beaver", "bear", "goose", "dog", "cat", "squirrel", "bird", "elephant", "horse",
 "bull", "giraffe", "seal", "bat", "skunk", "turtle", "whale", "rhino", "lion", "monkey",
 "frog", "alligator", "kangaroo", "hippo", "rabbit"};

int sequence[] = {0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0};     //start with an array full of 0s
```

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```
void setup() {

  pinMode(buttonPin, INPUT_PULLUP);       //set the button pin as an input

  lcd.init();
  lcd.backlight ();
  generateRandomOrder();                  //generate an array of random numbers from
1-25 that will determine which order the words are shown in

  showStartSequence();                    //print the start sequence text

}

void loop() {

  for(int i=0; i<arraySize; i++){              //for each of the 25 words in the sequence

     lcd.clear();                       //clear off the array

     roundNumber = i+1;                      //the array starts at 0, but the roundNumber will
start counting from 1
     lcd.print(roundNumber);                 //print the roundNumber (this is the current
round number)
     lcd.print(": ");                    //spacer between the number and the word
     lcd.print(words[sequence[i]]);        //print a random word from the word array

     startTime = millis();  //record the time that this round started

     while(digitalRead(buttonPin) == HIGH){          //do this until the button is
pressed...

       int roundedTime = round((timeLimit - (millis() - startTime))/1000);   //calculate the
time left in the round (dividing by 1000 converts the number to seconds
       lcd.setCursor(14,1);                                        //set the cursor in the lower
right corner of the screen
       lcd.print("  ");
```

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```
      lcd.setCursor(14,1);                                    //set the cursor in the lower
right corner of the screen
      lcd.print(roundedTime);                                  //print the time left in the
time limit
      delay(15);

      if(millis() - startTime > timeLimit){        //if the time limit is up before the button is
pressed
        gameOver();                          //end the game
      }

      if(digitalRead(buttonPin) == LOW){
         tone(buzzerPin, 272, 1);
      }

    }                  //exit this loop when the button is pressed

    delay(500);           //delay for a moment before giong on to the next ruond, so that
the button press doesn't get registered twice

   }
  //if you finish all 25 words
  winner();   //show the you win message
}


//--------------FUNCTIONS-----------------------------

//DISPLAYS A COUNTDOWN TO START THE GAME
void showStartSequence(){

  lcd.clear();               //clear the screen

  lcd.setCursor(0,0);          //move the cursor to the top left corner
  lcd.print("Category:");       //print "Category:"

  lcd.setCursor(0,1);          //move the cursor to the bottom left corner
  lcd.print("Animals");        //print "Animals:"
```

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```
  delay(2000);                //Wait 2 seconds

  lcd.clear();               //clear the screen
  lcd.print("Get ready!");      //print "Get ready!"
  delay(1000);                //wait 1 second

  lcd.clear();               //clear the screen
  lcd.print("3");             //print "3"
  delay(1000);                //wait 1 second

  lcd.clear();               //clear the screen
  lcd.print("2");             //print "3"
  delay(1000);                //wait 1 second

  lcd.clear();               //clear the screen
  lcd.print("1");             //print "3"
  delay(1000);                //wait 1 second
}

//GERNATES A RANDOM ORDER FOR THE WORDS TO BE DISPLAYED
void generateRandomOrder(){

  randomSeed(analogRead(0));           //reset the random seed (Arduino needs this to
generate truely random numbers

  for (int i = 0; i < 24; i++){        //do this until all 25 positions are filled

    int currentNumber = 0;             //variable to hold the current number
    boolean match = false;             //does the currentNumber match any of the
previous numbers?

    //generate random numbers until you've generated one that doesn't match any of
the other numbers in the arary
    do {
      currentNumber = random(0,arraySize);          //generate a random number from
1-25
```

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```
    match = false;                          //we haven't checked for matches yet, so start by
assuming that it doesn't match
    for (int i = 0; i < arraySize; i++){            //for all 25 numbers in the array
     if (currentNumber == sequence[i]){     //does the currentNumber match any of
the numbers?
       match = true;                        //if so, set the match variable to true
     }
    }
   } while(match == true);                  //if the match variable is true, generate another
random number and try again
   sequence[i] = currentNumber;             //if the match variable is false (the new
number is unique) then add it to the sequence
  }
}

//GAME OVER
void gameOver(){
  lcd.clear();                  //clear the screen

  lcd.setCursor(0,0);           //move the cursor the top left corner
  lcd.print("Game Over");       //print "Game Over"

  lcd.setCursor(0,1);           //move to the bottom row
  lcd.print("Score: ");         //print a label for the score
  lcd.print(roundNumber);       //print the score (the round number is the same as the
score)

  //play the losing fog horn
  tone(buzzerPin, 130, 250);     //E6
  delay(275);
  tone(buzzerPin, 73, 250);      //G6
  delay(275);
  tone(buzzerPin, 65, 150);      //E7
  delay(175);
  tone(buzzerPin, 98, 500);      //C7
  delay(500);

  while(true){}                 //get stuck in this loop forever
```

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```
}

//WINNER
void winner(){
  lcd.clear();                  //clear the screen

  lcd.setCursor(7,0);             //move the cursor to the top center of the screen
  lcd.print("YOU");               //print "You"

  lcd.setCursor(7,1);             //move the cursor to the bottom center of the screen
  lcd.print("WIN!");             //print "WIN!"

  //play the 1Up noise
  tone(buzzerPin, 1318, 150);     //E6
  delay(175);
  tone(buzzerPin, 1567, 150);     //G6
  delay(175);
  tone(buzzerPin, 2637, 150);     //E7
  delay(175);
  tone(buzzerPin, 2093, 150);     //C7
  delay(175);
  tone(buzzerPin, 2349, 150);     //D7
  delay(175);
  tone(buzzerPin, 3135, 500);     //G7
  delay(500);

  while(true){}                   //get stuck in this loop forever
}
```

5. Upload your program to the arduino. Select the Genuino Uno board, hook up your arduino board using the supplied USB cord to your computer and click “upload.”

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Your program is now loaded. Observe your LCD screen and complete your worksheet titled **Assignment 4D: "Who Am I?" Game.**

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Assignment 4D: “Who Am I?” Game

1. What happens if the time limit runs out while playing the game?

2. What happens when the player presses the button before the time limit is up? What is the greatest score a player can achieve?

3. What is the purpose of the liquid crystal library?

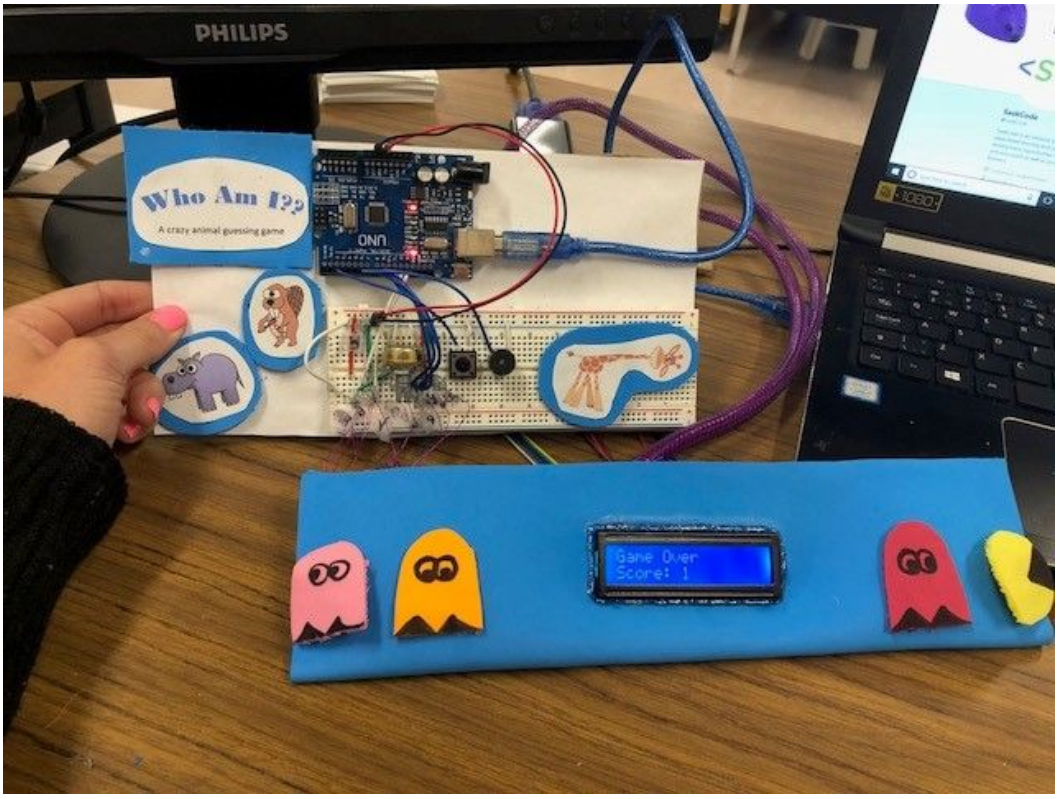
Extension Challenge:

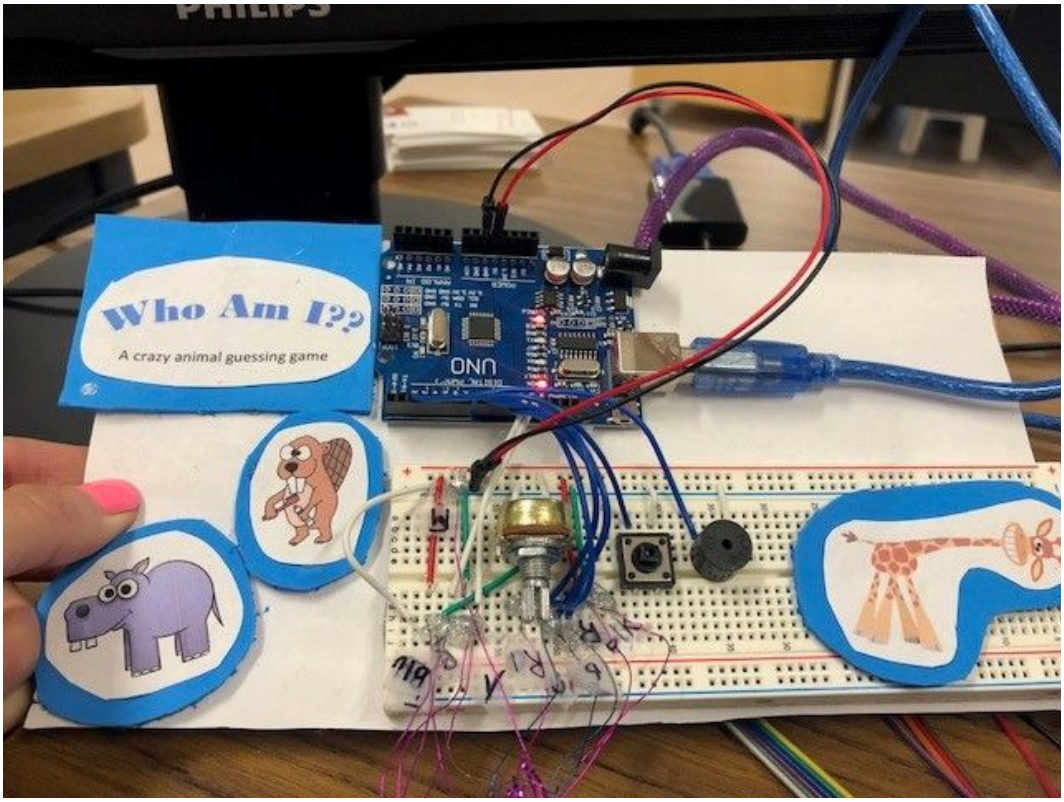
4. **CHANGE THE TIME LIMIT:** Changing the time limit variable will change the difficulty of the game.

5. **CHANGE THE WORDS IN THE WORD LIST:** Try changing the categories and words. The number of words in your words array must match the value of the variable arraySize .

6. **CHANGE THE WINNING AND LOSING SONGS**: By changing the tones in the winner() and gameover() functions you can change which song plays at the end of the game.

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